

Proof by induction - Divisibility

Basic skills

Q1. 3 divides 9

| symbol is read as divides

Q2. $f(n) = 5n - 4$

a) $f(1) = 1$ so $3 \nmid f(1)$

b) $f(2) = 6$ so $3 \mid f(2)$

c) $f(3) = 11$ so $3 \nmid f(3)$

d) $f(5) = 21$ so $3 \mid f(5)$

Q3. $f(n) = 3n + 9$

a) $f(1) = 12$ so $3 \mid f(1)$

b) $f(2) = 15$ so $3 \mid f(2)$

c) $f(3) = 18$ so $3 \mid f(3)$

d) $f(5) = 24$ so $3 \mid f(5)$

Q4. If $3 \mid f(n)$

$$g(n) = f(n) + 9$$

as $3 \mid f(n)$, $3 \mid 9$

we have $3 \mid f(n) + 9$

so $3 \mid g(n)$

more useful
in proof

Alternatively $f(n) = 3a$ for some a

$$\text{so } g(n) = 3a + 9 = 3(a+3)$$

so $3 \mid g(n)$

Q5. If $2 \mid f(n)$ $f(n) = 2a$ for some a

$$g(n) = f(n) + 2n^2 - 8$$

$$= 2a + 2n^2 - 8$$

$$= 2(a + n^2 - 4)$$

$= 2b$ for some b

so $2 \mid g(n)$

Competency skills

Q6. $f(n) = 11^n - 1$

a) $f(1) = 11^1 - 1 = 10$ so $5 \mid f(1)$

b) $f(k) = 5a$ for some a

so $11^k - 1 = 5a$

so $5a + 1 = 11^k$

c) $f(k+1) = 11^{k+1} - 1$
 $= 11^k \times 11 - 1$

so as $11^k = 5a + 1$

$$f(k+1) = (5a + 1) \times 11 - 1$$

$$= 55a + 11 - 1$$

$$= 55a + 10$$

$$= 5(11a + 2) \text{ so } 5 \mid f(k+1)$$

d) conclusion: As $5|f(1)$ and if $5|f(k)$ then $5|f(k+1)$ so $5|f(n)$ for all n

Q7. $f(n) = n^3 + 5n$

Base: $f(1) = 1^3 + 5 \times 1 = 6$ so $3|f(1)$

I.H : assume $3|f(k)$ for some k

i.e $f(k) = 3a$ for some a

so $k^3 + 5k = 3a$

now

$$f(k+1) = (k+1)^3 + 5(k+1)$$

$$= k^3 + 3k^2 + 3k + 1 + 5k + 5$$

$$= k^3 + 3k^2 + 8k + 6$$

find I.H $\swarrow$

$$= k^3 + 5k + 3(k^2 + k + 2)$$

$$= 3a + 3(k^2 + k + 2)$$

So $3 \mid f(k+1)$

Conclusion: As $3 \mid f(1)$ and if $3 \mid f(k)$ then $3 \mid f(k+1)$ we have by induction that $3 \mid f(n)$ for all n

Q8. $f(n) = 4^n + 6n - 1$

Base: $f(1) = 4^1 + 6 \times 1 - 1 = 9$

so $9 \mid f(1)$ holds

I.H: Assume $9 \mid f(k)$ for some k

i.e. $f(k) = 9a$ for some a

so $4^k + 6k - 1 = 9a$

$$4^k + 6k = 9a + 1$$

I.S: $f(k+1) = 4^{k+1} + 6(k+1) - 1$

$$= 4(4^k + 6k) - 18k + 6 - 1$$

find
I.H

$$= 4(9a+1) - 18k + 5$$

$$= 36a - 18k + 9$$

$$= 9(4a - 2k + 1)$$

$$\text{So } 9 \mid f(k)$$

Conclusion: As $9 \mid f(1)$ and if $9 \mid f(k)$ then $9 \mid f(k+1)$ we have by induction that $9 \mid f(n)$ for all n

Advanced skills

$$\text{Q 9. } (n+1)(n+3)$$

$$\text{Base case: } (1+1)(1+3) = 2 \times 4 = 8$$

$$4 \mid 8 \text{ so holds}$$

I.H : assume $4 \mid (k+1)(k+3)$ for some
 important! ~~to~~ odd k

$$\text{i.e. } (k+1)(k+3) = 4a \text{ for some } a$$

$$k^2 + 4k + 3 = 4a$$

I.S

next is $k+2$ if k is odd

$$(k+2+1)(k+2+3)$$

$$= (k+3)(k+5)$$

$$= k^2 + 8k + 15$$

$$= k^2 + 4k + 3 + 4k + 12$$

$$= 4a + 4k + 12$$

$$= 4(a+k+3)$$

$$\text{SO } 4 \mid (k+2+1)(k+2+3)$$

CONCLUSION: AS true for $n=1$ and
if true for odd k then true for
next odd no^o $(k+2)$ we have it
true for all odd numbers

Q10. $4^{t+1} + 5^{2t-1}$ Show divisible
by $1+1+2+3=7$

Base case: $t = 1$

$$4^2 + 5^1 = 16 + 5 = 21$$

7/21 so holds

I.H: Assume $7 \mid (4^{k+1} + 5^{2k-1})$ for
some k i.e. $4^{k+1} + 5^{2k-1} = 7a$
for some a

I.S:

$$\begin{aligned} & 4^{k+2} + 5^{2(k+1)-1} \\ &= 4^{k+2} + 5^{2k+1} \\ &= 4^{k+1} \times 4 + 25 \times 5^{2k-1} \\ &= 4(4^{k+1} + 5^{2k-1}) + 21 \times 5^{2k-1} \\ &= 4 \times 7a + 21 \times 5^{2k-1} \\ &= 7(4a + 3 \times 5^{2k-1}) = 7b \\ &\text{so } 7 \mid 4^{k+2} + 5^{2(k+1)-1} \end{aligned}$$

CONCLUSION! Hence as true for one car and if true for k cars then true for $k+1$ cars it is true for any no^o of cars.

Q11. $f(n) = 8^n + 1$

a) if $7 \mid f(k)$ $f(k) = 7a$ i.e. $8^k + 1 = 7a$

$$f(k+1) = 8^{k+1} + 1 = 8 \times 8^k + 1$$

$$= 8(8^k + 1) - 7$$

$$= 8 \times 7a - 7$$

$$= 7(8a - 1) = 7b \text{ so}$$

holds

b) Because missing base case, which isn't true as

$$7 \nmid f(1) \text{ as } f(1) = 9$$